

AMD Ryzen 7 8700G

An Odd Predicament

₹31,000



When AMD first launched the Ryzen processors with inbuilt graphics, they were targeting the segment of users interested in building entry-level DIY PCs and home theatre PCs. These configurations relied on an Integrated Graphics Processor for powering the display and running a few casual games. Anything heavier than that would require a discrete graphics card. Nevertheless, AMD's IGP in the 3200G was more powerful than what Intel had to offer with their desktop processors. Come Zen 3, we saw the 5000 series of Ryzen processors and this time around, AMD put out a mid-tier CPU in the form of the 5600G with inbuilt graphics. This too, did well as it was priced competitively and allowed for gamers to build a gaming PC for cheap with a temporary graphics solution until they could gather enough dough to get a discrete graphics card. We aren't mentioning the 4000G series



because those never entered the retail market. AMD tried selling them as a kit with an included motherboard in India but those didn't sell as expected. Now, with the 8000 series, AMD has yet again moved up one tier with the AMD Ryzen 7 8700G. With the past generations, the 'G' processors were simply rebadged mobile SoCs that could be marketed for the desktop segment. For example, the 5600G was a 'Cezanne' silicon processor built for laptops that was made into a desktop part. The 8700G is also a 'Phoenix' mobile chip that's being packaged for desktops. The exception is that the GPU has received a considerable bump in performance. With older 5000G series processors, the increased clocks on the desktop processors didn't translate to good gaming performance.

SPECIFICATIONS

'Phoenix' APUs include some of the first processors to include AMD's dedicated

AI accelerators called Ryzen AI. These NPUs are not present across the stack among the 8000G series but only in the 8600G and the 8700G. The 8500G and 8300G don't have the Ryzen AI NPUs. Instead, that place is utilised by Zen 4c cores. Zen 4c cores are basically the same as Zen 4 cores, it's just that the Zen 4c cores are physically smaller than Zen 4 cores and allow more denser packing. Each Zen 4c core is 35.4 per cent smaller than a Zen 4 Core and a 16-Core Zen 4c CCD (Core Complex Die) is barely 9.6 per cent larger than an 8-Core Zen 4 CCD. However, Zen 4c uses a slower cache so the IPC (Instructions Per Clock) is not the same as the normal Zen 4 cores. With the 8700G, this isn't an issue since it only has Zen 4 cores along with the Ryzen AI NPU.

The 8700G like most Ryzen 7 processors comes with 8 Cores / 16 Threads and the inbuilt GPU is a 780M based on RDNA 3. The GPU has 12 Compute Units with a total of 768 Shader Cores clocked at 2.9 GHz. The processor cores have a base clock of 4.2 GHz and under boost, they can go all the way up to 5.1 GHz. Compared to the 7700X, the L3 Cache on the 8000G processors is cut down by half to 16 MB. L3 Cache with the X3D processors has proven to be beneficial for accelerating gaming workloads so having a 'G' series processor that's designed for gaming with an L3 Cache that's less than half of its counterparts would lead to some interesting results.

Essentially, you've got the Ryzen 7 7700X which costs about INR 32K which has the same number of Cores and Threads but with 32 MB L3 Cache. Then there's the 8700G which is also priced at INR 32K with a better IGP but lower L3 Cache and there's always the 7800X3D which is currently priced at INR 37K. All there are competing for very similar set of customers and to make things a bit more confusing, there are rumours about AMD putting out an 8700F CPU which is the same as the 8700G but without the IGP. Obviously, we are all curious to see where each processor stands.